Enterprise AI Architecture Notes

Architecture Discussion (Preserved)

Before deciding architecture, here's what I currently believe the product is

The user opens a React application.

They can either:
• Ask natural language questions
• Generate reports

Natural language examples
• Which SKU performed best in Route 3 last quarter?
• Compare Route 2 and Route 5 during March.
• Which customers have declining purchases?

The AI understands the request.
If it's simply data retrieval, it converts it into SQL.
If calculations are needed, it calls deterministic Python code.
If charts are requested, it generates them.
If a report is requested, it generates PDF/Excel.

The LLM never calculates revenue, averages, growth, etc.
Python performs deterministic calculations.
The LLM simply decides which tools to use and explains the results.

PostgreSQL

Instead of Company PC → PostgreSQL, use AWS PostgreSQL (RDS) or AWS EC2 + PostgreSQL. Benefits: centralized backups, survives hardware failure, multiple AI servers can access it, easier future SaaS, no dependency on one office PC. Database should live in the cloud.

Excel Sync Service

Excel Folder/API → Sync Service → Validate → Transform → PostgreSQL. Keep it independent from the AI so it continues working even if AI is removed. Deploy beside PostgreSQL on AWS.

Python Business Logic

Store deterministic analytics in reusable modules such as growth.py, revenue.py, kpi.py, route_analysis.py, customer_analysis.py and forecast_helpers.py. These run on the same machine as FastAPI.

FastAPI

Acts as the backend brain exposing endpoints like /chat, /report, /chart, /upload, /auth and /admin. React never talks directly to the LLM. Runs on the AI server.

LangGraph

Orchestrates the workflow: Generate SQL? Call analytics? Generate report? Create chart? Summarize? Return answer? Keep it with FastAPI.

Python Sandbox

Used only for ad-hoc analysis not covered by coded analytics. Restricted with no internet, limited CPU/RAM, timeout and read-only datasets. Runs isolated on the AI server.

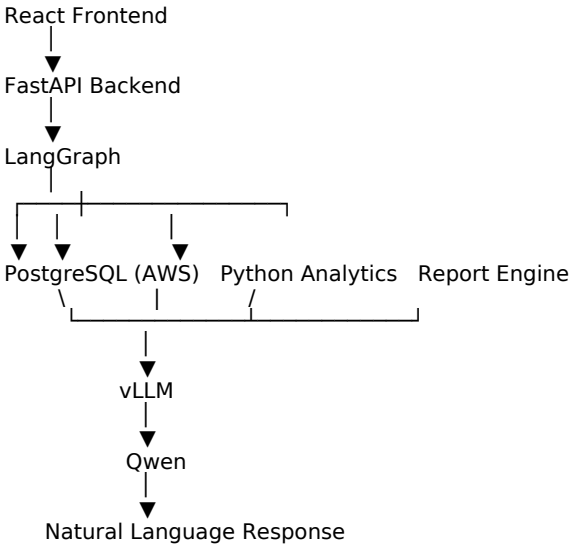
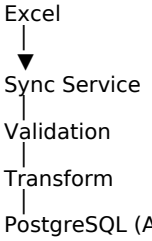
Qwen & vLLM

Run Qwen inside vLLM on the AI server together with FastAPI, LangGraph, Python analytics and the sandbox. AWS hosts PostgreSQL, Excel Sync Service and backups.

Frontend

Three experiences: AI Chat, Create Report (filters + instructions), and Dashboard for interactive exploration.

Overall Architecture



Separate Service: